Basic Boost Converter

Objective

This note examines the open-loop behavior of a DC–DC boost converter using software based simulation.
The focus is on:

- The relationship between duty cycle and average output voltage.
- The influence of inductance, capacitance, and switching frequency on ripple and startup transient response.

The goal is to characterize fundamental boost operation prior to introducing closed-loop control.

System Model

Input voltage:25VDC
Load:5Ω
Control method: Open-loop PWM
Duty ratios evaluated: D=0.3, 0.5, 0.7
Switching frequency:5kHz

Simulations were performed in PSIM to compare output-voltage and inductor-current waveforms under identical hardware conditions.

Basic Principle

A boost converter increases the input DC voltage by storing energy in an inductor and transferring that energy to the output.

When the switch is ON:
The inductor stores energy from the source, and the diode is reverse-biased, isolating the load.

$$V_{L,on} = V_{in}$$

When the switch is OFF:
The inductor's stored energy is released through the diode to the load, adding to the input voltage.

$$V_{L,off} = V_{in} - V_{out}$$

The average output voltage is given by:

$$V_{out} = \frac{V_{in}}{1 - D}$$

Example:
For $V_{in} = 25V$ and $D = 0.5$
$V_{out} \approx 50V$.

Circuit Parameters/Simulation Conditions

PARAMETER	SYMBOL	VALUE	DESCRIPTION
INPUT VOLTAGE	(Vin)	25V	Supply voltage
LOAD RESISTANCE	(R)	5Ω	Represents load
INDUCTOR	(L)	1mH	Controls current ripple
CAPACITOR	(C)	100μF	Smoothens output voltage
SWITCHING FREQUENCY	(fs)	5kHz	Determines ripple and loss
DUTY CYCLE	(D)	0.3 0.5 0.7	Target ratio for boost

These values were selected to maintain continuous conduction while allowing visible ripple and transient effects.

Simulation Results and Analysis

Effect of Duty Cycle on Output Voltage

The simulation results show that the average output voltage increases as the PWM duty cycle increases.

Duty Cycle	Theoretical Output Voltage	Simulated Output Voltage
0.3	35.71V	36V
0.5	50.00V	50V
0.7	83.33V	82V

The theoretical values are calculated using the ideal boost converter voltage gain equation.
The simulation results are close to the theoretical values. Small differences may result from simulation conditions, measurement methods, and the characteristics of the device models used.
The results confirm the expected relationship between duty cycle and output voltage. However, the output voltage does not increase linearly with duty cycle. As the duty cycle approaches 1, the theoretical voltage gain increases at a faster rate.

Steady-State Behavior

- The average output voltage increased nonlinearly with duty cycle.
- D=0.3 produced approximately 36V.
- D=0.5 produced approximately 50V.
- D=0.7 produced approximately 82V.

The nonlinear gain characteristic follows

$$V_{out} = \frac{V_{in}}{1 - D}$$

which becomes increasingly sensitive as D approaches 1.
Inductor current magnitude increased with duty ratio. Higher D resulted in larger peak and average inductor current.

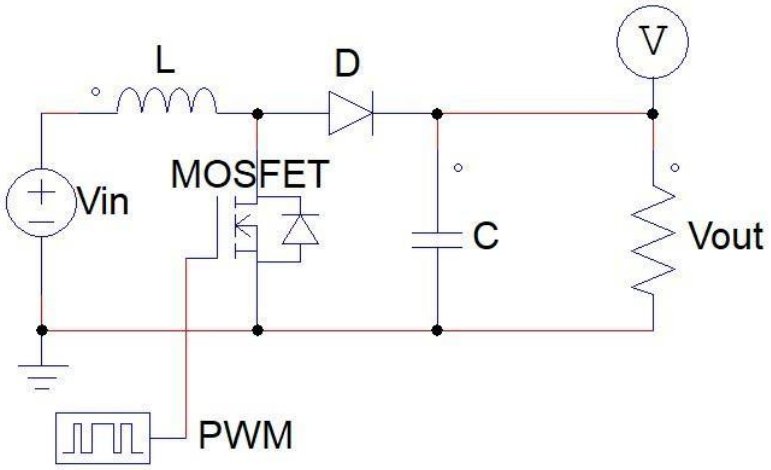


Figure 1

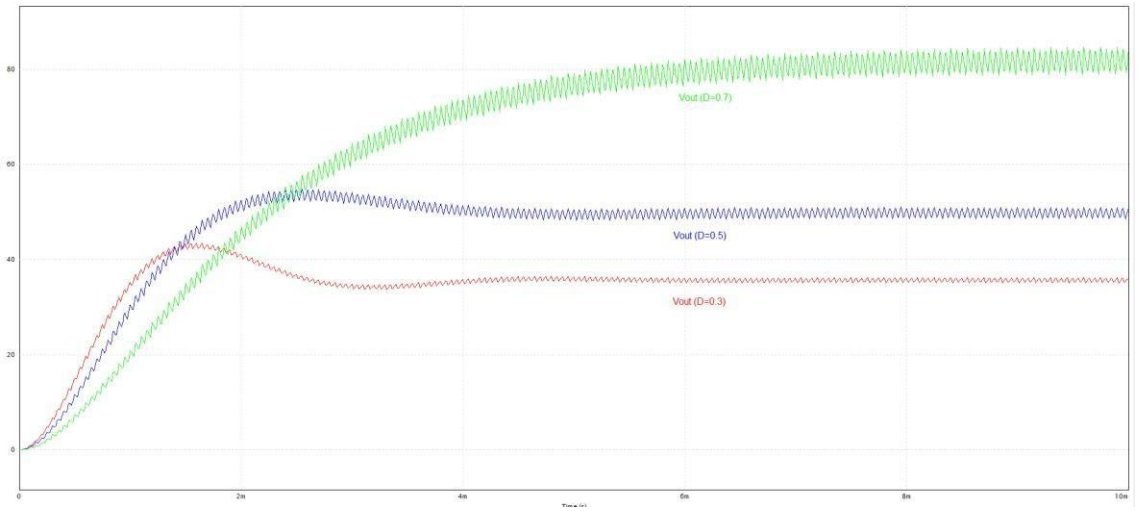


Figure 2

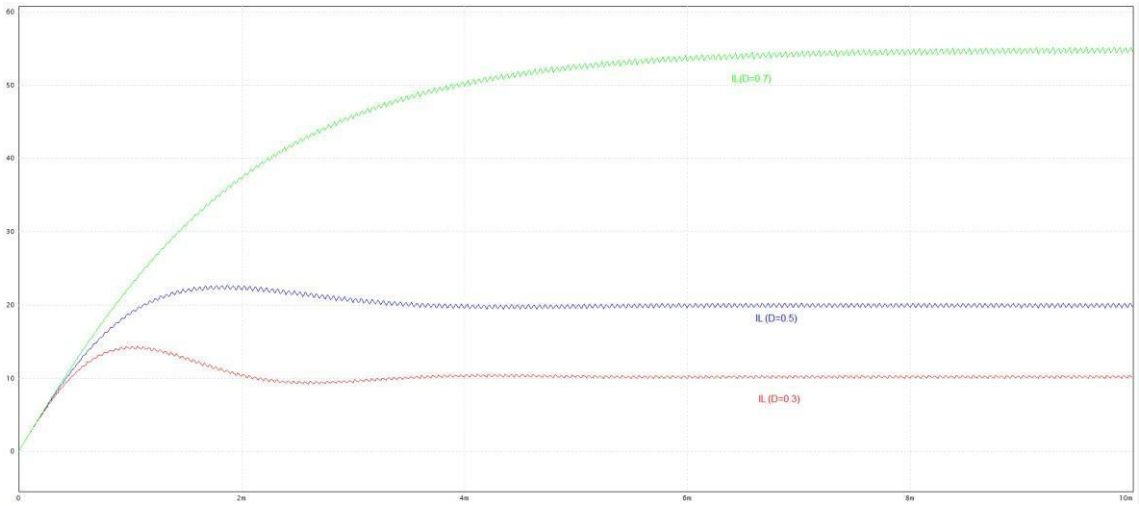


Figure 3

Using the simulation parameters, the theoretical inductor current ripple is:

Duty Cycle	Theoretical Inductor Current Ripple
0.3	1.5 A
0.5	2.5 A
0.7	3.5 A

Startup Behavior

- Voltage ripple remained moderate under nominal parameters.
- Rise time increased slightly at higher duty ratios.
- For D>0.6, transient overshoot became more pronounced due to increased stored inductor energy.

This indicates that high-duty operation increases both output gain and energy stress in the magnetic component.

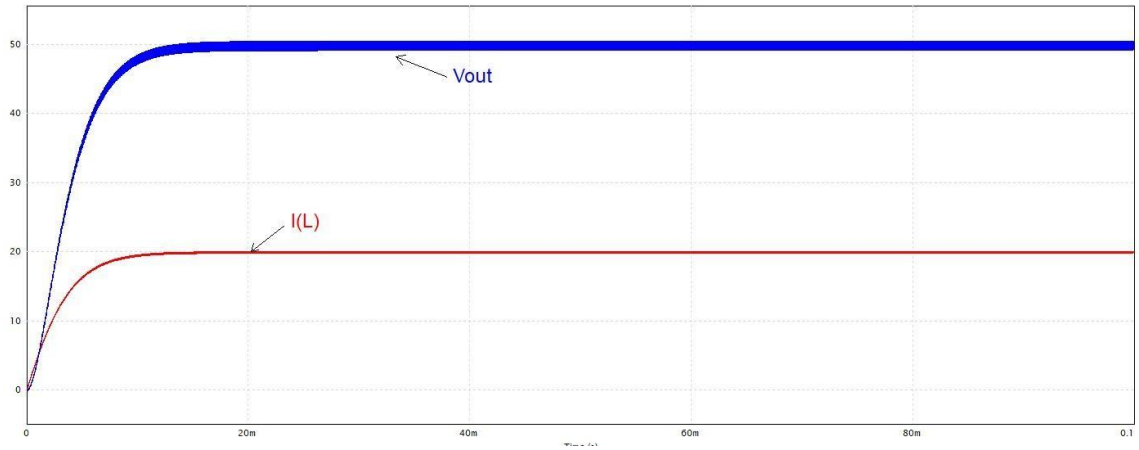


Figure 4

Parameter Sensitivity

Duty Cycle

- High duty ratios significantly increase inductor current and stress.
- Gain sensitivity increases as D approaches unity.

Inductance

- Larger L reduces current ripple.
- Larger L reduces overshoot but slows transient response.

Capacitance

- Larger C reduces output-voltage ripple.
- Excessive capacitance can delay settling if inductor energy release is large.

Switching Frequency

- Higher frequency reduces ripple magnitude.
- Higher frequency improves startup smoothness but may increase switching losses (not evaluated here).

Summary

- High duty ratios increase both output voltage and inductor current stress.
- Transient overshoot depends on stored magnetic energy and L, C selection.
- Boost converters become increasingly sensitive to duty variation at high D.

This establishes the open loop boost behavior baseline required before implementing feedback control.